JPE Replication Report

PUBLISHED
July 28, 2026

1 Paper Identification

ID	20220045
Author	Jinkins
Title	Two-sided Search in International Markets
Iteration	1

Some useful information up front:

- JPE Data and Code Availability Policy
- Step by step guidance from Data Editor for Authors
- Template README

2 Summary

In assessing compliance with our Data and Code Availability Policy, our reproducibility team has identified the following issues, which we ask you to address:

1. Your Data Availability Statement is incomplete. Please improve.
2. Instructions for how to execute the matlab code are insufficient.
3. We found some discrepancies in output.

3 General

- [x] Data Availability and Provenance Statements
  - [x] Statement about Rights - Part 1: Right to use the data (e.g. *did the authors lawfully obtain the data*)
  - [x] Statement about Rights - Part 2: Right to publish the data (e.g. *are human subjects involved and did permission been granted to publish their data?*)
  - [x] License for Data (optional, but recommended)
  - [ ] Details on each Data Source
- [ ] Dataset list
- [ ] Computational requirements
  - [x] Software Requirements
  - [ ] Controlled Randomness (as necessary)
  - [ ] Memory, Runtime, and Storage Requirements
- [x] Description of programs/code
  - [ ] License for Code (Optional, but recommended)
- [ ] {REPLICATOR} to Replicators
  - [ ] Details (as necessary)
- [ ] List of tables and programs
- [ ] References (Optional, but recommended)

Comments:

- Details on each data source are missing; please list all data sources separately and indicate whether they are publicly available. If they are publicly available, please state where and how they can be found. If the paper only uses statistics generated from a restricted dataset, please elaborate on this further. Also indicate if and where in the code the final datasets are prepared or if the code only uses pre-made datasets.
- The description and instructions for the code are rather short. Please elaborate more on the scripts that need to be run. Also provide more information on the different modes of the MATLAB script and the consequences of each mode. Provide clear instructions on how to run the code.
- Please provide an overview of all the tables and figures produced by the code, indicating which part of the code generates each output.

4 High-level Description of Research Project

This research project...

- [x] uses observational data.
- [x] uses *confidential* observational data.

- [ ] uses data generated via experiment or trial by the authors (in field or lab).
  - [ ] For lab experiments: The code to implement the experimental protocol (z-Tree, o-Tree etc) is included in the package.
  - [ ] A PDF copy of IRB approval is contained in the package.
  - [ ] A PDF document outlining the design of the experiment, including information on the selection and eligibility of participants is included.
  - [ ] A PDF copy of the instructions given to participants in both original language and an English translation is included.
- [ ] uses data generated via survey by the authors (in field or online)
  - [ ] The programs used to administer the survey (e.g. qualtrics survey file) are included.
  - [ ] A PDF copy of IRB approval is contained in the package.
  - [ ] A PDF of the original questionaire(s) and information on the selection and eligibility of participants is included.
- [ ] is a simulation study: the data are generated by an algorithm.
- [ ] is a numerical illustration: no data are used but code is used to produce illustrative output (tables, figures)

5 Data description

For any data that cannot be provided as part of the replication package, please provide the following information as part of the README.

> {REQUIRED} Please specify how long the data will be preserved in the restricted-access location.

> {REQUIRED} Please provide affirmation of support for replication checks. - As per the JPE policy, "In cases where data cannot be published in an openly accessible trusted data repository like the JPE dataverse, authors who have requested an exemption to publish them at the time of first submission must commit to preserving the data and code for a period of no less than five years following the publication of the paper. They should also provide reasonable assistance to requests for clarification and replication."

> {REQUIRED} Please provide a codebook for the data. - The codebook should at a minimum describe the variables obtained from the data source, in a manner that will let others verify that they have obtained a substantially similar dataset if successfully obtaining access to the data.

5.1 Data Sources

file/name	public	private	DOI/URL	Access described	cited readme	cited paper
U.S. Census LFTTD + microdata	no	yes	no	no	yes	yes
Census disclosure-approved outputs	no	yes	no	no	yes	yes
Annual Retail Trade Survey	yes	no	no	no	yes	yes
BEA apparel production	yes	no	no	no	yes	yes
WTO apparel trade	yes	no	no	no	yes	yes
Commerce apparel imports	yes	no	no	no	yes	yes
USITC Section 301 data	yes	no	no	no	yes	yes

6 Requirements

- [x] README is in TXT, MD, or PDF format
- [x] README is accessible at the root of the package (not inside a folder)
- [ ] Title conforms to guidance (starts with "Data and Code for: Title of Paper" or "Code for: Title of Paper", is properly capitalized)
- [x] Authors (with affiliations) are listed in the same order as on the paper

> {REQUIRED} Please review the title of the deposit as per our guidelines.

6.1 Stated Requirements

- [ ] No requirements specified

- [x] Software Requirements specified as follows:

- MATLAB R2025b with Global Optimization Toolbox and Parallel Computing Toolbox.
- Stata for Figure 11b, with SSC packages ebalance, ftools, require, and reghdfe.
- LibreOffice only if regenerating Word/Excel documentation files.

For missing requirements, see the list of required changes in Section 9

7 Analyzing Data and Code Files

7.1 Potential Personal Identifiable Information (PII)

⚠ We found the following instances of potentially personally identifying information. This may be completely legitimate but might be worth checking. *As a reminder, privacy legislation in many countries (e.g. GDPR in EU) prohibits the dissemination of personal identifiable information without prior (and documented) consent of individuals.* If indeed you want to publish such information with your replication package, you should probably have obtained IRB approval for this - please check!

Summary:

- Data files with PII indicators: 17
- Variables flagged in data: 25
- Code files with PII references: 162
- PII references in code: 1359

7.1.1 Summary of Flagged Files

File Type	File	Variables/References	PII Categories
Data	`AGS_sens_hetero.mat`	1	name
Data	`Optimization_2026_0209_0723_Run100of100.mat`	1	lat
Data	`Regression_hazard_rate_lnduration.csv`	1	lat
Data	`Regression_logLinear_Num_buyers_per_seller.csv`	1	lat
Data	`Regression_logLinear_Num_sellers_per_buyer.csv`	1	lat
Data	`Regression_logLinear_transaction_time_interval.csv`	1	lat
Data	`Regression_value_imp_duration_dur1.csv`	1	lat
Data	`merged_apparel_2017_2025_importVQT.dta`	2	country, lat
Data	`mp_model_appendix_table_parameters.csv`	1	city
Data	`se_results_baseline_no_M_target.mat`	1	name
Data	`se_results_baseline_no_M_target.mat`	1	name
Data	`se_results_est_eta.mat`	2	lon, name
Data	`se_results_est_eta.mat`	2	lon, name
Data	`se_results_hetero.mat`	1	name
Data	`se_results_no_M_target.mat`	1	name
Data	`table_09_mp_model_latex_comparison.csv`	6	lat
Data	`table_09_mp_model_source_parameters.csv`	1	city
Code	`00_do_apparel_shell.do`	7	country, lat, name
Code	`00_do_apparel_shell_abbrev.do`	2	country, lat
Code	`01_lfttd_imp_apparel_1996_2011.do`	59	name, country, lat, district

7.1.2 File Paths Report

Generated on 2026-07-10T13:44:11.772

Warning: Our search on file path types is imperfect and incurs both type 1 and type 2 errors. We aim to strike a reasonable balance between both. The below table is therefore only indicative. Detailed listings can be found in the appendix to this report.

In this table we analyse all files which contain source code (not data and documentation), and we report the grand total of each type of filepath we encountered.

Please check and replace any `windows` filepaths with unix compliant paths, insofar as this is possible in your setup. (Notice that `STATA` allows `/` to be a filepath separator on a windows platform, hence this is a requirement for *all* `STATA` applications.)

Files Analyzed	Windows Paths	Unix Paths	Mixed Paths	Drive Letters (`C:\` etc)
240	13	61	30	3

7.1.3 Duplicate Files Report

We found the following duplicate files:

Filename	Size (MB)	Checksum (MD5)
/code/matlab/baseline_counterfactuals_July_2024/baseline simulation/val_s.m	0.0	36ee8c88d3ccc7766814d980dac7a6077cab8893
/code/matlab/baseline_counterfactuals_July_2024/steady state summary/val_s.m	0.0	36ee8c88d3ccc7766814d980dac7a6077cab8893
/code/matlab/baseline_counterfactuals_July_2024/transition dynamics/val_s.m	0.0	36ee8c88d3ccc7766814d980dac7a6077cab8893
/code/matlab/baseline_counterfactuals_July_2024/trump tariff/val_s.m	0.0	36ee8c88d3ccc7766814d980dac7a6077cab8893
/code/matlab/baseline_no_NsNb_target/val_s.m	0.0	36ee8c88d3ccc7766814d980dac7a6077cab8893
/code/matlab/baseline_counterfactuals_July_2024/MP model/calibration_no_gamma_s/data_moments_concen.csv	0.0	a35e57482808a2fcb4c7755ee806bd76a7ca4c17
/code/matlab/baseline_counterfactuals_July_2024/baseline simulation/data_moments_concen.csv	0.0	a35e57482808a2fcb4c7755ee806bd76a7ca4c17
/code/matlab/baseline_counterfactuals_July_2024/steady state summary/data_moments_concen.csv	0.0	a35e57482808a2fcb4c7755ee806bd76a7ca4c17
/code/matlab/baseline_counterfactuals_July_2024/transition dynamics/data_moments_concen.csv	0.0	a35e57482808a2fcb4c7755ee806bd76a7ca4c17
/code/matlab/baseline_counterfactuals_July_2024/trump tariff/data_moments_concen.csv	0.0	a35e57482808a2fcb4c7755ee806bd76a7ca4c17
/code/matlab/baseline_no_NsNb_target/data_moments_concen.csv	0.0	a35e57482808a2fcb4c7755ee806bd76a7ca4c17
/code/matlab/heterogeneous_death_haz/sellers_only/data_moments_concen.csv	0.0	a35e57482808a2fcb4c7755ee806bd76a7ca4c17
/code/matlab/baseline_counterfactuals_July_2024/baseline simulation/intensity_s.m	0.0	295151a39d650d99ec2452a4f9c3cdaa8bbf92f9
/code/matlab/baseline_counterfactuals_July_2024/steady state summary/intensity_s.m	0.0	295151a39d650d99ec2452a4f9c3cdaa8bbf92f9
/code/matlab/baseline_counterfactuals_July_2024/transition dynamics/intensity_s.m	0.0	295151a39d650d99ec2452a4f9c3cdaa8bbf92f9
/code/matlab/baseline_counterfactuals_July_2024/trump tariff/intensity_s.m	0.0	295151a39d650d99ec2452a4f9c3cdaa8bbf92f9
/code/matlab/baseline_no_NsNb_target/intensity_s.m	0.0	295151a39d650d99ec2452a4f9c3cdaa8bbf92f9
/code/matlab/baseline_counterfactuals_July_2024/MP model/calibration_no_gamma_s/cum_cov.m	0.0	80e1e765162930aee7e5daeb148247f87ffd9ad9
/code/matlab/baseline_counterfactuals_July_2024/baseline simulation/cum_cov.m	0.0	80e1e765162930aee7e5daeb148247f87ffd9ad9
/code/matlab/baseline_counterfactuals_July_2024/steady state summary/cum_cov.m	0.0	80e1e765162930aee7e5daeb148247f87ffd9ad9
/code/matlab/baseline_counterfactuals_July_2024/transition dynamics/cum_cov.m	0.0	80e1e765162930aee7e5daeb148247f87ffd9ad9
/code/matlab/baseline_counterfactuals_July_2024/trump tariff/cum_cov.m	0.0	80e1e765162930aee7e5daeb148247f87ffd9ad9
/code/matlab/baseline_no_NsNb_target/cum_cov.m	0.0	80e1e765162930aee7e5daeb148247f87ffd9ad9
/code/matlab/heterogeneous_death_haz/sellers_only/cum_cov.m	0.0	80e1e765162930aee7e5daeb148247f87ffd9ad9
/code/matlab/mechanical_model/cum_cov.m	0.0	80e1e765162930aee7e5daeb148247f87ffd9ad9
/code/matlab/baseline_counterfactuals_July_2024/baseline simulation/define_states.m	0.0	a78ed4993492fc36e483822b91537f23b81d2142
/code/matlab/baseline_counterfactuals_July_2024/steady state summary/define_states.m	0.0	a78ed4993492fc36e483822b91537f23b81d2142
/code/matlab/baseline_counterfactuals_July_2024/transition dynamics/define_states.m	0.0	a78ed4993492fc36e483822b91537f23b81d2142

Filename	Size (MB)	Checksum (MD5)
/code/matlab/baseline_counterfactuals_July_2024/trump tariff/define_states.m	0.0	a78ed4993492fc36e483822b91537f23b81d2142
/code/matlab/baseline_counterfactuals_July_2024/trump tariff/val_st.m	0.0	afc48846fa8e9e5c0eebf9099485953ac2f8457e
/code/matlab/baseline_no_NsNb_target/define_states.m	0.0	a78ed4993492fc36e483822b91537f23b81d2142
/code/matlab/heterogeneous_death_haz/sellers_only/define_states.m	0.0	a78ed4993492fc36e483822b91537f23b81d2142
/code/matlab/mechanical_model/define_states.m	0.0	a78ed4993492fc36e483822b91537f23b81d2142
/code/matlab/baseline_counterfactuals_July_2024/MP model/calibration_no_gamma_s/variance_moments_concen.csv	0.0	a1c55e3842b333bcd5af315bb9b3a751b646402a
/code/matlab/baseline_counterfactuals_July_2024/baseline simulation/variance_moments_concen.csv	0.0	a1c55e3842b333bcd5af315bb9b3a751b646402a
/code/matlab/baseline_counterfactuals_July_2024/steady state summary/variance_moments_concen.csv	0.0	a1c55e3842b333bcd5af315bb9b3a751b646402a
/code/matlab/baseline_counterfactuals_July_2024/transition dynamics/variance_moments_concen.csv	0.0	a1c55e3842b333bcd5af315bb9b3a751b646402a
/code/matlab/baseline_counterfactuals_July_2024/trump tariff/variance_moments_concen.csv	0.0	a1c55e3842b333bcd5af315bb9b3a751b646402a
/code/matlab/baseline_no_NsNb_target/variance_moments_concen.csv	0.0	a1c55e3842b333bcd5af315bb9b3a751b646402a
/code/matlab/heterogeneous_death_haz/sellers_only/variance_moments_concen.csv	0.0	a1c55e3842b333bcd5af315bb9b3a751b646402a
/code/matlab/baseline_counterfactuals_July_2024/steady state summary/search_newbuyer.m	0.0	97d905cf7ee1c04dd695e8d0330aa6944a3126ec
/code/matlab/baseline_counterfactuals_July_2024/transition dynamics/search_newbuyer.m	0.0	97d905cf7ee1c04dd695e8d0330aa6944a3126ec
/code/matlab/baseline_counterfactuals_July_2024/trump tariff/search_newbuyer.m	0.0	97d905cf7ee1c04dd695e8d0330aa6944a3126ec
/code/matlab/baseline_no_NsNb_target/search_newbuyer.m	0.0	97d905cf7ee1c04dd695e8d0330aa6944a3126ec
/code/matlab/heterogeneous_death_haz/sellers_only/search_newbuyer.m	0.0	97d905cf7ee1c04dd695e8d0330aa6944a3126ec
/code/matlab/baseline_counterfactuals_July_2024/steady state summary/mom_function.m	0.0	f5b0b8bc3cefb84404f53bde81e63f1552902437
/code/matlab/baseline_counterfactuals_July_2024/transition dynamics/mom_function.m	0.0	f5b0b8bc3cefb84404f53bde81e63f1552902437
/code/matlab/baseline_counterfactuals_July_2024/trump tariff/mom_function.m	0.0	f5b0b8bc3cefb84404f53bde81e63f1552902437
/code/matlab/baseline_no_NsNb_target/mom_function.m	0.0	f5b0b8bc3cefb84404f53bde81e63f1552902437
/output/tables/paper/table_07_trump_section_301_tariff.tex	0.0	8db3e5f190f377e641b8adc23637456a0d8002bb
/output/tables/paper/table_09_mp_model_source_parameters.csv	0.0	3b28811321742fce2807331969653fbfe2c83995
/code/matlab/baseline_counterfactuals_July_2024/steady state summary/x2param.m	0.0	403886225fa76acd924070adbed33a8887f296ad
/code/matlab/baseline_counterfactuals_July_2024/transition dynamics/x2param.m	0.0	403886225fa76acd924070adbed33a8887f296ad
/code/matlab/baseline_no_NsNb_target/x2param.m	0.0	403886225fa76acd924070adbed33a8887f296ad
/code/matlab/baseline_counterfactuals_July_2024/trump tariff/val_bt.m	0.0	77a0b3e186af65603f9ec42e60be8d4cbc5e8a8a
/code/matlab/baseline_counterfactuals_July_2024/steady state summary/intensity_b.m	0.0	1e913885f3f66cc6d76faa5b05aa406f89cc9552
/code/matlab/baseline_counterfactuals_July_2024/transition dynamics/intensity_b.m	0.0	1e913885f3f66cc6d76faa5b05aa406f89cc9552
/code/matlab/baseline_counterfactuals_July_2024/trump tariff/intensity_b.m	0.0	1e913885f3f66cc6d76faa5b05aa406f89cc9552
/code/matlab/baseline_no_NsNb_target/intensity_b.m	0.0	1e913885f3f66cc6d76faa5b05aa406f89cc9552
/code/matlab/baseline_counterfactuals_July_2024/baseline simulation/val_b.m	0.0	18f0f900ea31b5284192c56432efc7eadd80ed64
/code/matlab/baseline_counterfactuals_July_2024/steady state summary/val_b.m	0.0	18f0f900ea31b5284192c56432efc7eadd80ed64
/code/matlab/baseline_counterfactuals_July_2024/transition dynamics/val_b.m	0.0	18f0f900ea31b5284192c56432efc7eadd80ed64
/code/matlab/baseline_counterfactuals_July_2024/trump tariff/val_b.m	0.0	18f0f900ea31b5284192c56432efc7eadd80ed64

Filename	Size (MB)	Checksum (MD5)
/code/matlab/baseline_counterfactuals_July_2024/trump tariff/define_parameters.m	0.0	97d97f5ad76fce56a548925a33b747739fa9154b
/output/tables/paper/table_06_market_developments_counterfactual.tex	0.0	072dc637ae6a3808f6d675a17f8bd77364da47c8
/code/matlab/baseline_counterfactuals_July_2024/steady state summary/eval_s.m	0.0	4073b4b2067abf14e499037cd994a3fc559b0b40
/code/matlab/baseline_counterfactuals_July_2024/transition dynamics/eval_s.m	0.0	4073b4b2067abf14e499037cd994a3fc559b0b40
/code/matlab/baseline_counterfactuals_July_2024/trump tariff/eval_s.m	0.0	4073b4b2067abf14e499037cd994a3fc559b0b40
/code/matlab/baseline_no_NsNb_target/eval_s.m	0.0	4073b4b2067abf14e499037cd994a3fc559b0b40
/code/matlab/heterogeneous_death_haz/sellers_only/eval_s.m	0.0	4073b4b2067abf14e499037cd994a3fc559b0b40
/code/matlab/baseline_counterfactuals_July_2024/baseline simulation/define_lindex.m	0.0	7fc3b761bc2c982c493f8018fe273053c017195f
/code/matlab/baseline_counterfactuals_July_2024/steady state summary/define_lindex.m	0.0	7fc3b761bc2c982c493f8018fe273053c017195f
/code/matlab/baseline_counterfactuals_July_2024/transition dynamics/define_lindex.m	0.0	7fc3b761bc2c982c493f8018fe273053c017195f
/code/matlab/baseline_counterfactuals_July_2024/trump tariff/define_lindex.m	0.0	7fc3b761bc2c982c493f8018fe273053c017195f
/code/matlab/baseline_no_NsNb_target/define_lindex.m	0.0	7fc3b761bc2c982c493f8018fe273053c017195f
/code/matlab/heterogeneous_death_haz/sellers_only/define_lindex.m	0.0	7fc3b761bc2c982c493f8018fe273053c017195f
/code/matlab/baseline_counterfactuals_July_2024/steady state summary/transition_s.m	0.0	e7888b10ac1fc89e35e4d635716473429c4875be
/code/matlab/baseline_counterfactuals_July_2024/transition dynamics/transition_s.m	0.0	e7888b10ac1fc89e35e4d635716473429c4875be
/code/matlab/baseline_counterfactuals_July_2024/trump tariff/transition_s.m	0.0	e7888b10ac1fc89e35e4d635716473429c4875be
/code/matlab/baseline_no_NsNb_target/transition_s.m	0.0	e7888b10ac1fc89e35e4d635716473429c4875be
/code/matlab/baseline_counterfactuals_July_2024/trump tariff/objective.m	0.0	1c58c7bba6aa276dfca2d645cd185688c26827c3
/code/matlab/baseline_counterfactuals_July_2024/trump tariff/plot_fit.m	0.0	bdf6a5a24db55bff45d8c0bfb2e6767fea96a572
/code/matlab/baseline_no_NsNb_target/TSS_mutationgaussian.m	0.0	f214d4ea90e0e2b6c6456d4c7bbe208e2cfde1b3
/code/matlab/heterogeneous_death_haz/sellers_only/TSS_mutationgaussian.m	0.0	f214d4ea90e0e2b6c6456d4c7bbe208e2cfde1b3
/code/matlab/baseline_counterfactuals_July_2024/steady state summary/define_payoffs.m	0.0	3803094365c0f284fddc4154244ee6f303cb6485
/code/matlab/baseline_counterfactuals_July_2024/transition dynamics/define_payoffs.m	0.0	3803094365c0f284fddc4154244ee6f303cb6485
/code/matlab/baseline_no_NsNb_target/define_payoffs.m	0.0	3803094365c0f284fddc4154244ee6f303cb6485
/code/matlab/heterogeneous_death_haz/sellers_only/define_payoffs.m	0.0	3803094365c0f284fddc4154244ee6f303cb6485
/code/matlab/baseline_counterfactuals_July_2024/steady state summary/sim_moments_baseline.m	0.01	9b92b8f524d6ad0411a7b010caf32ed5ee30c1a4
/code/matlab/baseline_counterfactuals_July_2024/transition dynamics/sim_moments_baseline.m	0.01	9b92b8f524d6ad0411a7b010caf32ed5ee30c1a4
/code/matlab/baseline_counterfactuals_July_2024/trump tariff/sim_moments_baseline.m	0.01	9b92b8f524d6ad0411a7b010caf32ed5ee30c1a4
/output/tables/paper/table_03_partner_count_distribution_source.xls	0.01	d3d6c8746b6fbad04485c7a1b398cf89e3ff0450
/code/matlab/baseline_counterfactuals_July_2024/transition dynamics/data_moments_baseline.m	0.01	38dbaa8455bebdc94e32dca1b0aff013c28b930e
/code/matlab/baseline_counterfactuals_July_2024/trump tariff/data_moments_baseline.m	0.01	38dbaa8455bebdc94e32dca1b0aff013c28b930e
/output/tables/paper/table_01_transition_buyers_per_supplier.xls	0.01	cda1099c9941bd95447abf7ab565fd203ab7cfc8
/output/tables/paper/table_02_transition_suppliers_per_buyer.xls	0.01	edf8f6849a4e4590e0f6d20f7b0bf26b47e779cd
/output/figures/paper/figure_02_number_of_suppliers.eps	0.01	3d0410b2dbf6fa25375c774e3154e306b788a79f
/output/figures/paper/figure_06_buyers_related_arm_length.eps	0.01	c42a6722af1aa1b92966e12a248524bee893bed1
/output/figures/paper/figure_03_suppliers_by_country.eps	0.02	4de4ef52d3540cb62b15ea63b199b181b5146b69
/code/matlab/heterogeneous_death_haz/sellers_only/05_avg_number_partners_by_yr_in_market.xls	0.02	354b1607de0d565100dcf2b18a517880bb8d04cf

Filename	Size (MB)	Checksum (MD5)
/code/matlab/baseline_no_NsNb_target/results/se_results_no_M_target.mat	1.88	47868283bc8243def11870d73c6b23abcc56955b
/code/matlab/mechanical_model/results/se_results_baseline_no_M_target.mat	1.88	47868283bc8243def11870d73c6b23abcc56955b

7.1.4 Zero Size Files Report

We did not find any zero sized files.

7.1.5 Large Files Report

We found the following files larger than 100MB:

Filename	Size (MB)
/code/matlab/baseline_counterfactuals_July_2024/trump tariff/dynamics_tariff.mat	140.3
/data/US_ITC_figure11b_event_study/merged_apparel_2017_2025_importVQT.dta	310.7

7.2 Code description

There is one master-matlab file and one separate Stata do-file. As said earlier there is no overview in the README that explains which parts of the scripts produce specific figures or tables.

cloc **github.com/AlDanial/cloc v 2.06 T=0.16 s (1206.3 files/s, 145797.1 lines/s)**

Language	files	blank	comment	code
MATLAB	143	3348	2899	11552
TeX	5	252	87	2447
Stata	14	340	334	1662
CSV	33	91	0	758
Text	2	8	0	32
——–	——–	——–	——–	——–
SUM:	197	4039	3320	16451

- [x] The replication package contains a “main” or “master” file(s) which calls all other auxiliary programs.

> NOTE: In-text numbers that reference numbers in tables do not need to be listed. Only in-text numbers that correspond to no table or figure need to be listed.

7.3 Computing Environment of the Replicator

- Nuvolos cloud 4vCPU & 16GB RAM
- Stata/MP 19.5
- Matlab R2025b

8 Replication

- Downloaded code and data from dropbox
- Opened MATLAB per README
- Used the ’fast’track for this replication
- ran the master-script and additional stata do-file per README

9 Findings

- Running `run_all_replication.m` initially failed with a MATLAB licensing error when the code tried to launch a separate MATLAB process via `system('matlab -batch ...')`. I believe this was not a bug in the replication code, but an environment issue of Nuvolos: my session already had a valid license, but spawning a new process this way doesn't seem to inherit it, so the new process couldn't find one.

Fix: replaced the external `system()` call with `eval()`, running the script directly within the already-licensed session instead of starting a new process. This resolved the error:

```
function runMatlabScript(scriptDir, scriptName)
oldDir = pwd;
cd(scriptDir);
try
    runIsolated(scriptName);
catch ME
    cd(oldDir);
    error('MATLAB generator failed: %s\n%s', scriptName, ME.message);
end
cd(oldDir);
end

function runIsolated(scriptName)
eval(scriptName);
end
```

- Did not encounter any further issues related to code itself, everything seemed to run smoothly.

9.1 Missing Requirements

- [ ] Software Requirements
  - [x] Stata
    - [x] Version
    - Packages
- [x] Computational Requirements specified as follows:
  - Cluster size, disk size, memory size, etc.

> [REQUIRED] Please amend README to contain complete requirements.

9.2 Data Preparation Code

No raw data preparation code is included in this replication package. The authors have already processed the confidential data and provided the derived statistics needed for replication. The only code that needs to be run for this replication is the master script (run_all_replication.m) in MATLAB,this ran smoothly.

9.3 Tables and Figures

- Most of the generated tables and figures match the output in the paper. However, there are two where the output shows some discrepancies. Example:
- Figure 4: Graph is completely different compared to the one in the paper.

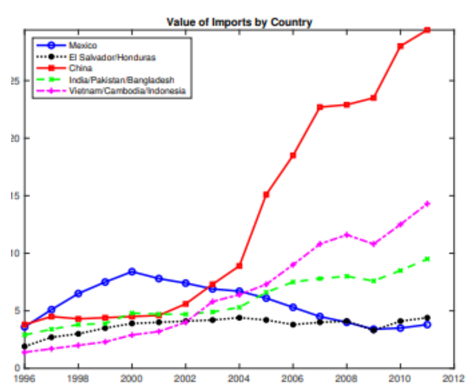
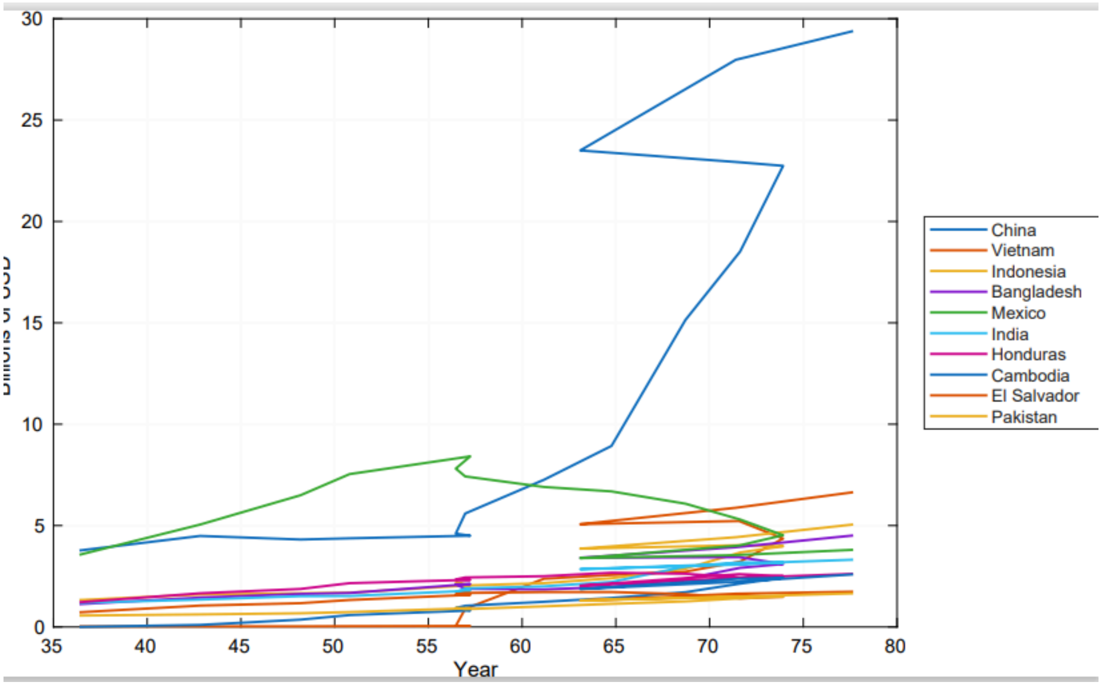


Figure 4: Value of imports by country, 1996-2011[c]

Paper version



Replicated version

- Table 13: I can't make the connection (only a few coefficients) between the generated output and the table in the paper. Please elaborate further.

metric	pre_policy	post_policy	pct_change	ejtx_policy_
measure a	0.1232709	0.1399155	135.024.265.896.116	29.3
measure a	0.0245451	0.0298434	215.858.982.575.767	19.3
measure a	0.1478161	0.1697590	148.447.044.535.228	NaN
share high-	0.1660521	0.1757991	586.983.425.673.058	NaN
average ex	######	######	-129.258.911.577.682	NaN
consumer v	0.2833541	0.2989868	551.703.806.834.417	NaN

Replicated version 1

metric	ejtx_policy_pct_change	aar_policy_pct_change	
measure active low-p	292.763.157.894.737	135.024.265.896.116	
measure active high-	192.898.272.552.783	215.858.982.575.767	
measure active buye	-0.942587832047992	NaN	
total profit	low-productivity suppli	248.746.238.716.149	NaN
total profit	high-productivity supp	-649.087.221.095.335	NaN
total profit	buyers	0.596421471172959	NaN
total search costs	low-productivity suppli	240.759.240.759.241	NaN
total search costs	high-productivity supp	-855.769.230.769.231	NaN
total search costs	buyers	117.370.892.018.779	NaN
number of suppliers	592.991.913.746.632	NaN	
high-productivity sup	-165.622.202.327.663	NaN	
consumer welfare	-0.833333333333341	551.703.806.834.416	
derived: measure ac	279.337.835.553.301	148.447.044.535.228	
derived: high-type sh	-675.658.614.936.015	586.983.425.673.058	

Replicated version 2

Table 13: Section 6.1 experiment: EJTX vs. our AAR model (% changes)

Metric	EJTX*	our AAR model
Measure active low-quality suppliers	+29.3	+13.5
Measure active high-quality suppliers	+19.3	+21.6
Measure active buyers	−0.9	—
High-quality suppliers per buyer	−16.6	—
Consumer welfare $(1/P)$	−0.8	+5.5

Paper version

9.4 Other Issues

- The code and replication itself seem almost ready to go, but the README needs further work.

10 Classification

- [ ] full reproduction
- [ ] full reproduction with minor issues

- [x] partial reproduction (see above)
- [ ] not able to reproduce most or all of the results (reasons see above)

10.1 Reason for incomplete reproducibility

> {REPLICATOR}: mark the reasons here why full reproduciblity was not achieved, and enter this information in new issue on the github repo of this package. When results are fully reproduced, leave this section here, and mark "None".

- [ ] None.
- [x] `Discrepancy in output` (either figures or numbers in tables or text differ)
- [ ] `Bugs in code` that were fixable by the replicator (but should be fixed in the final deposit)
- [ ] `Code missing`, in particular if it prevented the replicator from completing the reproducibility check
  - [ ] `Data preparation code missing` should be checked if the code missing seems to be data preparation code
- [ ] `Code not functional` is more severe than a simple bug: it prevented the replicator from completing the reproducibility check
- [ ] `Software not available to replicator` may happen for a variety of reasons, but in particular (a) when the software is commercial, and the replicator does not have access to a licensed copy, or (b) the software is open-source, but a specific version required to conduct the reproducibility check is not available.
- [ ] `Insufficient time available to replicator` is applicable when (a) running the code would take weeks or more (b) running the code might take less time if sufficient compute resources were to be brought to bear, but no such resources can be accessed in a timely fashion (c) the replication package is very complex, and following all (manual and scripted) steps would take too long.
- [ ] `Data missing` is marked when data *should* be available, but was erroneously not provided, or is not accessible via the procedures described in the replication package
- [ ] `Data not available` is marked when data requires additional access steps, for instance purchase or application procedure.
- [ ] `Missing README` is marked if there is no README to guide the replicator, or the README is not in compliance with JPE requirements

10.2 Potential Personal Identifiable Information (PII)

⚠ We found the following instances of potentially personally identifying information. This may be completely legitimate but might be worth checking. *As a reminder, privacy legislation in many countries (e.g. GDPR in EU) prohibits the dissemination of personal identifiable information without prior (and documented) consent of individuals.* If indeed you want to publish such information with your replication package, you should probably have obtained IRB approval for this - please check!

Summary:

- Data files with PII indicators: 17
- Variables flagged in data: 25
- Code files with PII references: 162
- PII references in code: 1359

10.2.1 Summary of Flagged Files

File Type	File	Variables/References	PII Categories
Data	`AGS_sens_hetero.mat`	1	name
Data	`Optimization_2026_0209_0723_Run100of100.mat`	1	lat
Data	`Regression_hazard_rate_lnduration.csv`	1	lat
Data	`Regression_logLinear_Num_buyers_per_seller.csv`	1	lat
Data	`Regression_logLinear_Num_sellers_per_buyer.csv`	1	lat
Data	`Regression_logLinear_transaction_time_interval.csv`	1	lat
Data	`Regression_value_imp_duration_dur1.csv`	1	lat
Data	`merged_apparel_2017_2025_importVQT.dta`	2	country, lat
Data	`mp_model_appendix_table_parameters.csv`	1	city
Data	`se_results_baseline_no_M_target.mat`	1	name
Data	`se_results_baseline_no_M_target.mat`	1	name
Data	`se_results_est_eta.mat`	2	lon, name
Data	`se_results_est_eta.mat`	2	lon, name

File Type	File	Variables/References	PII Categories
Data	`se_results_hetero.mat`	1	name
Data	`se_results_no_M_target.mat`	1	name
Data	`table_09_mp_model_latex_comparison.csv`	6	lat
Data	`table_09_mp_model_source_parameters.csv`	1	city
Code	`00_do_apparel_shell.do`	7	country, lat, name
Code	`00_do_apparel_shell_abbrev.do`	2	country, lat
Code	`01_lfttd_imp_apparel_1996_2011.do`	59	name, country, lat, district